PROJECT

PROMETHEUS

Quantitative Economics & Neural Finance

Full System Blueprint, Research Protocol & Deployment Gates

ECONOSIM-X	Multi-agent computational macroeconomics
STOCHASTICA	Neural SDEs and deep stochastic control
VOLTERRA	Derivatives, stochastic volatility, neural PDEs
TITAN	Institutional-style risk and portfolio analytics

MASTER SPECIFICATION | VERSION 1.0 | 02 NOVEMBER 2026

Design documentation complete. Implementation, verification, and approvals NOT claimed.

Research / simulation only. Not financial advice or permission for live trading.

Document control and integrity statement

What is completed today?
This master specification PDF is the completed deliverable. All software features, experiments, security checks, and operational controls are NOT VERIFIED unless independent evidence is attached later. Do not mark research or deployment items complete without evidence.

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Project type: Open-source research and simulation platform; proposed future enterprise research deployment. Default operating mode: offline analysis and paper simulation only.

Owner: Project maintainers (to be appointed). Status: Design specification published; no code, dataset, trained model, independent model validation or production control is certified by this document.

Status key	Meaning	Required evidence
[ ] NOT VERIFIED	Requirement documented; implementation or test not demonstrated.	Issue/PR + test/log/report + reviewer.
[x] VERIFIED	Work implemented and supported by accepted evidence.	Evidence artifact, owner, timestamp and approver.
[!] BLOCKED	Cannot proceed until a risk or dependency is resolved.	Blocker ticket and approved mitigation.
[~] N/A	Not applicable for specific deployment scope.	Signed justification and scope owner.

Decision-grade scope distinction

Maturity gate	Permitted usage	Mandatory boundary
R0 - Experimental	Local simulation, benchmark, offline research	Synthetic or appropriately licensed data; no live orders.
R1 - Reproducible research	Internal reproducible studies and published research artifacts	Validated experiments, data provenance and scientific limitations.
R2 - Enterprise research service	Permissioned internal analytics and scenario tooling	Security, monitoring, incident response, independent review, approved SLOs.
R3 - Regulated decision / live trading	Only if a separate institution authorizes each deployment and activity	Jurisdiction-specific legal, operational, supervisory and model validation approvals; outside initial product scope.

Regulatory frameworks referenced below are contextual sources, not statements that the project or its developers are regulated entities. The April 2026 US interagency model risk guidance supersedes SR 11-7; it is the contemporary model risk reference where relevant [R11].

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01 Executive concept and mission

PROMETHEUS is a modular scientific computing platform for investigating feedback between heterogeneous economic agents, stochastic financial markets, derivative valuation, neural hedging, and portfolio risk. Its purpose is falsifiable economic and financial research, not automated speculative trading or purported replication of proprietary systems.

Primary scientific objective

The central hypothesis
Explicit macro-financial feedback may improve out-of-sample calibration and robustness of neural risk-control policies relative to comparable market-only baselines under unseen economic shocks. The result can be positive, negative, or inconclusive; no improvement is assumed.

Target users and decisions

Persona	Main activity	Output
Computational economist	Design economic agents, policies and macro shocks	Balance-sheet-consistent scenario reports
Quant researcher	Calibrate stochastic markets and price derivatives	Risk-neutral pricing and model error diagnostics
ML researcher	Learn neural dynamics and hedging policies	Training artifacts, ablations, OOD reports
Risk analyst	Analyze scenario tail losses and portfolio constraints	VaR/ES dashboards, stress testing, evidence
Platform / model risk reviewer	Reproduce outputs, inspect assumptions and controls	Approval evidence and signed gate reports

In scope

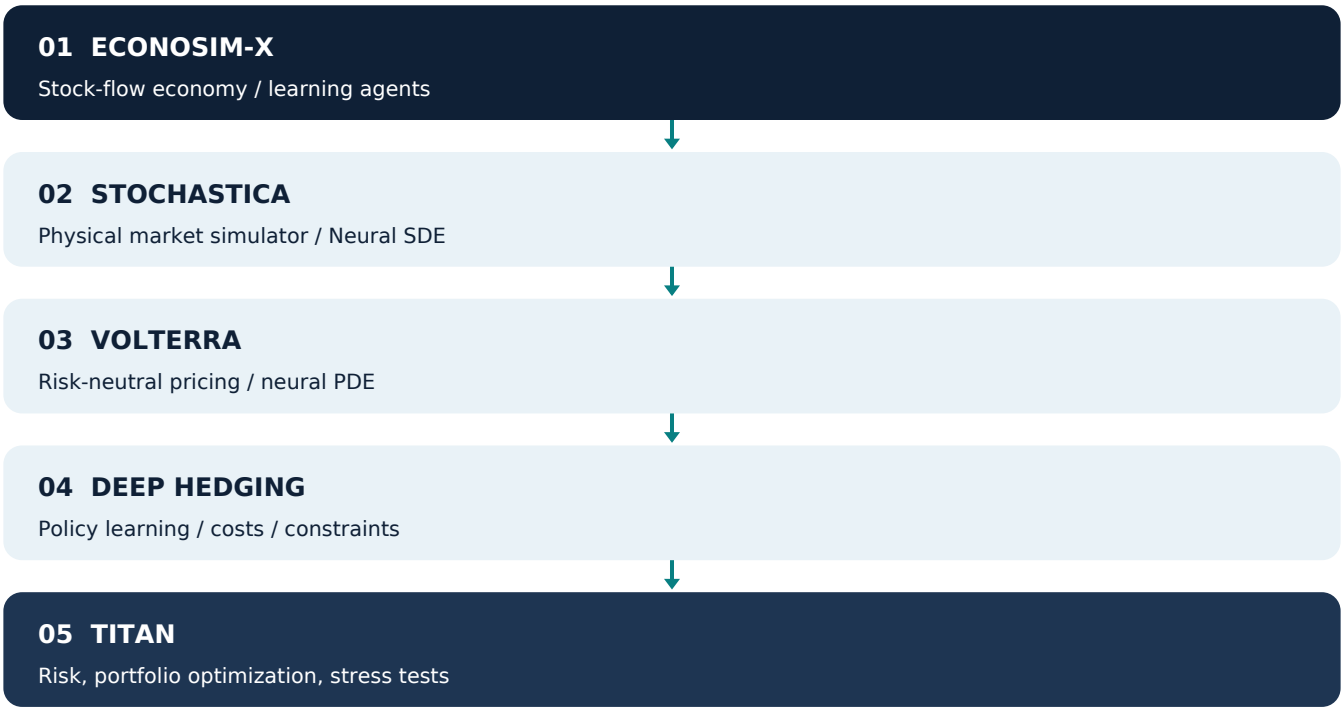
- Simulation of households, firms, banks and monetary-policy agents with a consistent ledger.
- Classical and learned physical-measure financial dynamics; separately specified pricing-measure derivatives models.
- Derivative pricing, sensitivities, neural hedging under costs, tail-risk estimation, optimization and stress testing.
- Point-in-time economic datasets, versioned experiments, audit trails, reproducibility and controlled research-serving APIs.

Explicitly out of scope at initial launch

- Live order routing, brokerage connectivity, custody, real-money execution and automated regulated investment advice.
- Claims that simulated economies identify causal effects in real economies without additional identification.
- Black-box overrides of market-clearing/accounting rules; synthetic performance marketed as historical results.
- Certification against any financial regulation or security standard by document authorship alone.

02 End-to-end architecture and contracts

The minimum viable scientific architecture is a modular monolith with typed subsystem interfaces and an append-only experiment event log. Service decomposition should follow measured operational needs, not fashion.



Feedback decisions enter the NEXT economic step; no same-tick lookahead.

Control-plane versus data-plane

Plane	Responsibilities	Contract
Simulation plane	Macroeconomic event clock, agent actions, clearing, market paths	Deterministic replay from frozen config and RNG states.
Model plane	Classical model calibration, neural training, model registry	Versioned model package with training-data fingerprint.
Risk plane	Pricing, Greeks, positions, P&L, risk, optimization	Position and market snapshots share as-of timestamp.
Control plane	Identity, authorization, approvals, audit, monitoring	No unapproved transition across deployment stages.
Evidence plane	Datasets, test logs, validations, run manifests, reports	Tamper-evident provenance with immutable artifact IDs.

Event timing and causal order

Suggested clock: one macroeconomic step per month; a nested market calendar advances across actual business days. Every macro data point has both observation and publication time. Financial trades and portfolios feed into the next eligible economic state rather than leaking their future outcomes into current decisions.

1. Release information: Make only available economic releases and revisions visible.
2. Apply external shocks: Inject exogenous policy, productivity, liquidity or disaster shocks.
3. Compute actions: Agents choose household, firm, credit and policy actions from observed state.

4. Clear and settle: Apply budget constraints, balance-sheet journal entries and market clearing.
5. Simulate financial step: Draw physical-measure shocks and update relevant financial states.
6. Value and hedge: Run risk-neutral valuation, compute hedge actions with transaction costs.
7. Compute portfolio risk: Assess realized P&L, conditional losses, constraints and limits.
8. Commit next state: Snapshot event ledger, audit hash and state for subsequent step.

Integration contract

Contract	Required fields	Invariant
EconomicState	as_of, known_at, macro vector, agent ledger hash, shock ID	No feature publication later than known_at.
MarketState	as_of, scenario ID, P/Q measure tag, S, r, v, correlation	No pricing from unlabeled measure.
DerivativeBook	instrument ID, payoff, maturity, quantity, pricing config	Valuation inputs synchronized to market snapshot.
HedgeAction	policy ID, observation time, target holdings, constraints	Action cannot depend on future returns.
RiskReport	portfolio version, model ID, confidence level, horizon	Risk metric definitions and units present.
RunManifest	code digest, data hash, seed, dependency lock, result hash	Every benchmark reproduces from manifest.

03 Mathematics and economic identities

3.1 Economic agent learning

Economic state transition

```
z[t+1] = F(z[t], actions[1..N,t], exogenous_shock[t+1])
```

F must enforce accounting identities and constraints; do not use an unrestricted neural transition as a substitute for economic consistency.

Agent decision rule

```
a[i,t] = policy_theta_i(observation[i,t]); a[i,t] in feasible_set[i,t]
```

Partial observability must be explicit. All actions are projected or optimized over feasible sets.

Bellman equation

```
V_i(s) = max_a { r_i(s,a) + gamma * E[ V_i(s_next) | s,a ] }
```

This is a design abstraction. Heterogeneous agents may optimize different objectives and information sets.

3.2 Stock-flow consistency

At every event, transaction journals must preserve double entry. For a stylized loan origination, the lending bank increases its loan asset and its matching deposit liability by the same amount. Subsequent payments transfer deposits, reserves or claims in a properly specified settlement model. Initial macro calibration must enforce the relevant sectoral budget identities.

Balance-sheet identity

```
Assets[sector,t] = Liabilities[sector,t] + Equity[sector,t]
```

Test tolerance: zero for integer-currency ledger operations; numerical tolerance pre-approved for floating aggregations.

Intertemporal budget constraint (stylized)

```
A[t+1] = (1+r[t])A[t] + income[t] - consumption[t] - taxes[t]
```

Units, timing, price deflation and borrowing constraints must be specified for every agent class.

3.3 Physical versus pricing measures

Physical measure: asset returns

```
dS/S = mu_theta(z,x) dt + sigma_theta(z,x) dW^(P)
```

Calibrate expected returns and risk dynamics under historical/physical measure P.

Risk-neutral valuation

```
V[t] = E^(Q) [ exp(-Integral_t^T r(s)ds) * payoff(S[T]) | F[t] ]
```

Do not equate physical-measure neural forecasts with risk-neutral prices; calibrate Q dynamics or specify a pricing kernel.

3.4 Volatility, hedging, and risk

Heston reference dynamics

```
dS = mu*S*dt + sqrt(v)*S*dW_s ; dv = kappa*(theta-v)dt + xi*sqrt(v)*dW_v
```

Brownian increments are correlated with corr(dW_s,dW_v)=rho. Choose a positivity-preserving numerical scheme.

Neural policy under hedging friction

```
h[t] = policy_phi(state[t], positions[t]) ; Cost[t] = c * abs(h[t]-h[t-1])
```

Add spread, market impact, limits and funding in later stages; do not imply frictionless execution.

Loss-tail objective

```
min_phi CVaR_alpha(L[phi]) + lambda * E[turnover(phi)]
```

Specify whether transaction costs are already included in L to avoid double-counting penalties.

Expected shortfall (continuous-tail version)

```
ES_alpha(L) = E[L | L >= VaR_alpha(L)]
```

For discrete distributions/ties, implement the quantile-integral definition with an explicit estimator.

3.5 Derivative price numerical consistency

Black-Scholes PDE reference

```
V_t + 0.5*sigma^2*S^2*V_SS + r*S*V_S - r*V = 0
```

Boundary and terminal conditions are part of the benchmark, not optional metadata.

For PINN or deep-BSDE models, compare numerical price and Delta/Gamma against independent benchmark solvers; enforce terminal payoff, non-negativity where appropriate, monotonicity and static-arbitrage constraints. Neural approximation is not automatically arbitrage-free. Key research references: Heston [R02], Han/Jentzen/E [R03] and Buehler et al. [R01].

04 ECONOSIM-X: computational economy

Agent and sector specification

Agent / market	State variables	Decision / model
Households	Cash, deposits, debt, labor, income, wealth, expectations	Consumption, savings, labor participation, borrowing, default.
Firms	Capital, productivity, wages, inventories, debt	Employment, production, prices, capex, financing.
Commercial banks	Assets, deposits, loans, reserves, capital, nonperforming loans	Underwriting, pricing credit, collateral, lending limits.
Central bank	Policy rate, inflation/output targets, balance sheet	Rule-based Taylor-style baseline; constrained learning only later.
Government	Tax receipts, transfers, spending, debt	Fixed fiscal rules first; optional policy experiments.
Markets	Goods, labor, credit, funding and financial assets	Clearing, rationing, settlement, prices, liquidity.

Baseline economic dynamics

Start from analytical/behavioral baselines: household consumption out of available income, firm production with capacity limits, a central-bank policy rule, bank capitalization and credit rationing. Introduce multi-agent PPO only after the rules-based economy passes conservation and stylized-fact tests. Learning agents must not invent resources, create money outside allowed banking mechanisms, or spend unavailable capital.

Economic validity tests

- Market-clearing residuals, economy-wide balances, conservation and wealth-flow checks.
- Impulse response experiments: monetary tightening, productivity improvement, credit rationing, tax/transfer changes.
- Sensitivity analysis to heterogeneous expectations, risk aversion, policy coefficients and shock persistence.
- Benchmark empirical moments: inflation persistence, output volatility, credit cyclicality, correlations and lag patterns.
- Ablate learned decisions: rules-based, single-agent learning, full MARL; do not assume emergent results imply empirical realism.

Handling central bank RL responsibly

A learned central-bank controller should be used as an experimental counterfactual policy mechanism, not a recommended real policy. Define social welfare and constraints transparently, and show whether the learned policy depends on unrealistic information, exploitable reward definitions, or non-robust simulation assumptions.

05 STOCHASTICA: stochastic market dynamics

Model family ladder

Tier	Models	What must be established
0 - Analytical	GBM, Bachelier/normal reference where appropriate	Known distributions, reproducible path construction.
1 - Classical	Heston, GARCH, correlated diffusion, regime switching	Calibration fit, positivity, covariance validity.
2 - Neural	Neural drift/diffusion, neural stochastic volatility	Stable discretization, non-explosive paths, distribution fit.
3 - Macro conditioned	Neural SDE with macroeconomic state and publishing lag	No leakage from future releases, improvement against Tier 1.
4 - Stress robust	Domain-randomized and distributionally robust policies	Performance on held-out structural shocks.

Numerics and calibration

Use correlated Brownian draws with a positive semidefinite covariance matrix. Compare Euler-Maruyama and suitable positivity-oriented Heston schemes, perform strong/weak convergence tests where relevant, and inspect negative variance and exploding trajectories. Distinguish training under P from pricing calibration under Q.

Necessary diagnostics

- Return distribution: skew, excess kurtosis, conditional tails, volatility clustering and leverage effect.
- Stochastic volatility: Feller-condition region, variance positivity under discretization, stability vs step size.
- Cross-asset dependence: correlations, tail dependence, changing covariance regime, positive semidefinite projections.
- Neural dynamics: gradient stability, solver sensitivities, parameter identifiability and train/holdout gap.
- Macro link: conditioned-on-release-time versus intentionally leaked features as a negative control.

06 VOLTERRA: pricing and numerical finance

Derivative coverage

Product	Baseline	Advanced extension
European call/put	Black-Scholes closed-form; put-call parity	Heston Fourier pricing, neural PDE.
Digital call	Analytical/finite-difference comparison	Near-discontinuity solver error study.
Barrier call/put	Monte Carlo with barrier monitoring correction	PDE and path-dependent neural extension.
Asian arithmetic option	Monte Carlo + control variate	Neural path encoder.
Vanilla portfolios	Aggregated Greeks and risk	Surface-consistent recalibration and hedging.

Pricing architecture

The engine takes a versioned instrument specification, market snapshot, risk-neutral model, solver, and tolerance budget. It returns price, Greeks, convergence diagnostics, calibration timestamp and model version. Never report a lone price without method, units, error estimate and as-of date.

Arbitrage and numerical diagnostics

- Put-call parity under consistent rates/dividends; discount bounds and positive option value.
- For vanilla calls: sensible monotonicity in spot, strike and maturity under applicable assumptions.
- Volatility surface calendar and butterfly spread checks using documented interpolation assumptions.
- Monte Carlo path-count convergence, standard errors, independent seed set, antithetic/control-variate diagnostics.
- PINN/BSDE terminal-boundary residuals plus held-out price and Greeks error against a trusted solver.

07 Neural hedging and stochastic control

Hedging policy families

Family	Observation	Baseline / objective
Delta baseline	Spot, time, implied volatility / calibrated model	Black-Scholes or model-consistent delta rehedge.
No-trade band	Greeks, transaction costs, position	Cost-aware threshold policy.
MLP	Observable market state + current holdings	Terminal hedging-loss optimization.
GRU / LSTM	Causal sequence of observable states	Path-dependent control with bounded history.
Actor-critic	State, action, realized transition and reward	Constrained sequential control; reward audit.
Robust policy	Distribution of stress regimes / uncertainty	Worst-case/CVaR-aware hedge selection.

Objective and failure modes

Hedging objectives must specify whether risk is measured on terminal discounted wealth, running P&L or liquidity-adjusted losses. Penalize constraint violations explicitly; a model beating delta hedging by using more leverage is not a fair comparison. Include fees, bid/ask spreads, execution latency and market impact progressively.

Experimental splitting

Separate independent simulation seeds for training, validation and test. Also use held-out parameter regimes and model families: train on GBM/Heston and evaluate under perturbed volatility, jumps or altered spreads when specified. Never tune on the final test scenarios. Report uncertainty across training seeds AND evaluation scenarios.

08 TITAN: institutional-style risk and portfolio analytics

Risk feature specification

Risk domain	Metrics	Evidence standard
Market risk	Historical/parametric VaR, ES, volatility, drawdown	Exception count, ES scoring, horizon/quantile labels.
Factor risk	Macro, rates, equities and style-factor exposures	Reconciliation to asset and portfolio returns.
Liquidity risk	Turnover, spread, participation and liquidation assumptions	Transparent assumptions, conservative stress tests.
Credit/systemic	Bank default cascades and propagation networks	Synthetic bilateral exposures clearly labeled.
Stress testing	Historical scenarios and generated macro shocks	Shock provenance, scenario plausibility, replay.
Optimization	Mean-variance, CVaR, robust/turnover-constrained	Feasibility, dual diagnostics, transaction-cost impact.

Estimation guardrails

Use an explicit horizon, confidence level, observation window, mark-to-market convention and P&L mapping. Avoid calling a simplified ES number regulatory Expected Shortfall. Any Basel-style or other regulatory benchmark is for educational comparison and needs jurisdiction-/product-specific interpretation [R13].

Portfolio optimization requirements

Investment constraints include budget, no-short/short bounds, turnover, exposure caps, leverage, liquidity and trading costs. For convex formulations, verify solver optimality/feasibility; for nonconvex extensions, report local-solution sensitivity. Backtest with time-varying estimated covariance and trading frictions.

09 Data, vintages, schemas and licensing

Proposed datasets

Source	Fields / role	Governance note
FRED-MD/QD [R04]	Monthly/quarterly macro factors and economic states	Read metadata; normalize release and transformation details.
ALFRED [R05]	Revised and originally released series values	Store vintage/known-at timestamps for reconstruction.
US Treasury [R06]	Reference Treasury yield curve	Record rates, conventions, business-day calendar.
Kenneth French [R07]	Asset-pricing portfolio factors	Track updated dataset methodology and licensing.
Licensed market-data vendor	Asset prices, corporate actions, options surfaces	Explicit contract, entitlements and redistribution rights.
Synthetic generators	Balance sheets, interbank obligations, stress paths	Always label synthetic, preserve source config and seed.

Point-in-time data schema

```
series_id: string; observation_at: timestamp; published_at: timestamp; vintage_at: timestamp
ingested_at: timestamp; asof_eligible_at: timestamp; unit: string; frequency: enum
value: decimal/float; source_url: string; source_license: string; raw_hash: SHA256
revision_id: string; transformation_id: string; timezone: string; quality_flags: list
```

Point-in-time rule

Strict data inclusion predicate

For a forecast issued at time T, include an observation only when published_at <= T and access to that particular vintage is independently documented. Missing or revised series must be handled without retroactive interpolation leakage.

Data partitioning

Separate raw immutable snapshots, normalized canonical tables, feature tables, simulation output, model checkpoints and public derivative artifacts. Hash each layer and retain upstream dependency IDs. Never commit large licensed datasets, credentials or private records to the public repository.

10 Scientific evaluation and experiment matrix

Primary and secondary hypotheses

H1: Macro-conditioned neural hedging improves held-out risk-score performance vs cost-aware baseline under exogenous macro shocks. H2: Any improvement remains after point-in-time release restrictions. H3: Gains persist under distribution shift, or the model correctly signals uncertainty. H4: Neural pricing approximators achieve a predefined accuracy/latency trade-off without systematic arbitrage violations.

Factorial ablation matrix

Factor	Reference setting	Experimental setting
Economic simulation	Rules-based macro shock generator	Stock-flow-consistent learning agents
Market dynamics	Heston or regime-switching classic model	Macro-conditioned neural SDE
Hedging	Cost-aware delta / no-trade band	MLP, GRU or constrained RL
Pricing	Analytical/Fourier/Monte Carlo reference	PINN or deep-BSDE approximation
Evaluation regime	In-distribution synthetic paths	Held-out structural shock families
Data vintage	Release-time compliant data	Leaked-data negative-control diagnostic only

The core 2 × 2 × 2 economy × market × hedge factorial design yields eight configurations. Pricing approximation and shifted-regime experiments are second-stage experiments; avoid exploding to every possible cross-product without a pre-registered plan.

Metrics, controls and tests

Research object	Metrics	Disqualifying failure
Economy	Accounting residual, market imbalance, moments, impulse response	Ledger nonconservation or impossible agent balance.
Market SDE	Distribution scores, tail coverage, stability, calibration	Future feature access or numerical explosion.
Pricing	Abs/relative pricing error, Greek error, arbitrage checks	Material pricing/parity violations.
Hedging	CVaR, mean P&L, cost, turnover, constraint breaches	Leverage unfairness or uncosted execution.
Risk forecast	VaR exception, ES joint/consistent score, coverage, calibration	Unlabeled horizon or invalid P&L basis.
Platform	Determinism, run hash, latency, availability, audit completeness	Nonreproducible promoted result.

Statistics protocol

Predefine primary endpoint, effect direction, alpha, multiple-comparison policy, resampling unit, and minimum economically meaningful effect. For overlapping horizons, apply appropriate dependent-data inference (e.g. block bootstrap). Report confidence intervals, paired comparison design, seed variability, negative controls and failed experiments.

11 Platform engineering and API specifications

Implementation architecture

Begin with a Python package and local DuckDB/Parquet research store. Add Postgres for metadata, object storage for immutable artifacts, Redis only if a proven queue/cache requirement emerges, and an API gateway for R2. Docker images and reproducible environments should be pinned by digest. Prefer modular monolith boundaries to premature distributed services.

Service interfaces (proposed)

Endpoint	Input contract	Response / safeguards
POST /v1/simulations	Config ID, scenario, seed, max steps	Async job ID; validated schema and resource quota.
GET /v1/simulations/{id}	Authorized experiment ID	State, deterministic manifest and audit metadata.
POST /v1/pricing	Instrument book, market snapshot, model ID	Prices, Greeks, numerical diagnostics; no execution.
POST /v1/risk/reports	Portfolio snapshot, scenario/horizon	Report artifact ID, traceable model and metric definitions.
POST /v1/models/evaluate	Registered candidate, approved benchmark suite	Review report and gate recommendation; no promotion.
POST /v1/models/promote	Two-person approval, validation evidence	Immutable decision with separation-of-duty checks.
GET /health and /ready	Service and dependency probes	Strict readiness vs liveness distinction.

Schema design

Define versioned Pydantic/protobuf models for EconomicState, AgentLedger, MacroObservation, MarketScenario, PricingRequest, PricingResult, HedgeDecision, PositionBook, RiskReport, ExperimentManifest, ModelCard. Include UTC timestamps, explicit units, decimal/currency convention, confidence/horizon, and backward-compatible schema versions. Reject unknown or malformed monetary units.

Repository, CI and documentation

Use a src/ layout, pyproject.toml with dependency lock, typed public APIs, unit/integration/property tests, Sphinx/MkDocs documentation, GitHub Actions for CI, pre-commit, semantic versions and data/model provenance. Never use test-set performance to select a production model.

12 MLOps, reliability, security and governance

Training-to-serving model lifecycle

Candidate -> trained -> evaluated -> independent review -> approved -> staged -> canary/shadow -> promoted -> monitored -> retired. Each transition requires a signed immutable decision, artifact checksum, configuration, dataset fingerprint, benchmark scores, and rollback handle. Separation of duties is mandatory for R2+

Security baseline

- Strong identity and least privilege: SSO/OIDC and service identities at R2.
- Secrets vault, short-lived tokens, TLS in transit, encrypted storage, restricted artifact access.
- Dependency scanning, software bill of materials, pinned base images, vulnerability remediation SLAs.
- Audit immutability, access logs, anomaly alerts, protected branches and signed release tags.
- Threat model for malicious datasets, poisoned checkpoints, prompt-free API misuse, model extraction and denial of service.
- Role-based restrictions on sensitive portfolio data and approvals for any vendor data egress.

Operational reliability assumptions: proposals, not verified SLAs

SLO / objective	Proposed R2 threshold	Measurement design
Availability	99.5% monthly for internal research API	Synthetic probe and independent service metric.
API read p95	Under 500 ms for small metadata reads	Documented workload and baseline hardware.
Job creation p95	Under 2 s for validated simulation requests	Excludes queued scientific compute time.
Audit completeness	100% of model promotions and risk-report generation	Reconcile operation count to immutable logs.
Recovery point objective	24 h maximum for research metadata; lower if approved	Restore drill plus timestamp evidence.
Recovery time objective	4 h for internal research APIs	Timed full restoration exercise.
Numerical drift	Within approved precision-specific tolerances	Reference fixture CPU/GPU comparisons.

Model risk and controls as of November 2026

Use the Federal Reserve/OCC/FDIC Revised Guidance on Model Risk Management dated 17 April 2026 as a model-risk reference, noting that the guidance itself is tailored to institutions and not automatic legal applicability to this research tool [R11]. NIST AI RMF supplies voluntary Govern/Map/Measure/Manage concepts [R08]. NIST SSDF informs secure development [R09]. BCBS 239 supplies relevant risk-data traceability and reporting principles [R12]. For EU regulated financial entities, DORA is applicable from January 2025, but applicability must be assessed by counsel [R14].

Hard deployment boundary

No implicit permission for live trading
PROMETHEUS R0-R2 produces simulations, reports and offline recommendations only. Any future trading connection must be a separate authorized product with legal review, appropriate market-access controls, pre-trade limits, broker agreements, human oversight, staged testing and operational sign-off. The SEC market-access rule is a relevant reference for covered US broker-dealers, not a blanket requirement for all deployments [R15].

13 Phased engineering program and exit gates

Gate	Deliverable	Non-negotiable exit criteria
G0: Scientific design freeze	Research plan, mathematical assumptions, interface schemas and data strategy	Signed spec and open-risk inventory; no untested claim of economic realism.
G1: Ledger-correct economic kernel	Rules-based agents, double-entry, settlement and shock engine	100% accounting invariants; deterministic replay and fault tests.
G2: Classical market and pricing benchmarks	GBM/Heston simulation, Black-Scholes/Fourier/MC reference	Price/Greek benchmarks and numerical convergence reports pass.
G3: First deep hedging experiment	MLP hedge, cost-aware baseline, CVaR reporting	Held-out seeds, reproducibility, paired uncertainty report.
G4: Neural stochastic and economics learning	Neural SDE, learning households/firms, OOD shocks	Demonstrated positive/rejected hypotheses and failure diagnostics.
G5: Integrated research release	Risk engine, full ablations, portfolio and stress experiments	Independent scientific reproduction, documented limitations.
G6: Enterprise research hardening	SSO, versioned APIs, logging, CI/CD, SLO, DR, validation	R2 controls tested with owner-specific evidence and sign-offs.
G7: Regulated deployment assessment (optional)	Institution-specific legal/compliance/market-access work	Separate scope and governance decision; never implied by G6.

Indicative workload and compute

No fixed completion date can be responsibly guaranteed. Complexity drivers are agent dynamics, realistic data, calibration, independent model validation, and deployment controls. CPU-first baseline development should precede GPU training; small PyTorch agents and path simulation can be prototyped on a single mid-range GPU, while large MARL populations, nested stochastic optimization and stress-grid sweeps may require cloud/HPC resources. Record GPU memory, energy/runtime and experiment cost for every benchmark.

Research artifacts per milestone

Each gate produces an ADR (architecture decision record), specification diff, reproducibility manifest, scenario dataset hash, evaluation summary, failure log, code commit/release, identified reviewer, risk-acceptance decisions and a gate decision (GO / CONDITIONAL / NO-GO).

14 Comprehensive implementation and readiness checklists

The following master checklist is intentionally extensive. Each line defines a distinct capability and its minimum acceptance evidence. Every implementation checkbox starts unchecked because no source code, infrastructure or production validation was supplied for verification. Completing the PDF does not complete the software.

Checklist completion protocol
To set [x], link an immutable artifact (PR, commit, report, dataset hash, CI run, screenshot, independent approval), add owner and date to the issue tracker, and obtain appropriate review. A checklist is not a certification of regulatory compliance. Evidence examples below are proposed gates, not proof of execution.

GOV | Program governance and scope (14 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
GOV-01 [ ] NOT VERIFIED	Publish mission, user personas, permitted actions, non-goals	Approved charter with signed owner and scope version.
GOV-02 [ ] NOT VERIFIED	Define R0/R1/R2/R3 release classes and promotion authority	Gate policy covers paper-only default and live-trading exclusion.
GOV-03 [ ] NOT VERIFIED	Assign accountable technical, scientific, security and risk owners	RACI register with named owners and deputies.
GOV-04 [ ] NOT VERIFIED	Maintain requirements-to-tests traceability	Each P0 requirement links to executable test and result.
GOV-05 [ ] NOT VERIFIED	Create architecture decision record process	At least one reviewed ADR per major scientific interface.
GOV-06 [ ] NOT VERIFIED	Document conflicts of interest and intended audience	Public disclosure reviewed for claims and financial interests.
GOV-07 [ ] NOT VERIFIED	Define research ethics and simulation limitations	Approved limitations statement in documentation and reports.
GOV-08 [ ] NOT VERIFIED	Create issue prioritization P0/P1/P2 policy	Backlog triaged with severity, owner, milestone and SLA.
GOV-09 [ ] NOT VERIFIED	Identify third-party intellectual property and attribution needs	License inventory and notices approved by maintainer.
GOV-10 [ ] NOT VERIFIED	Document jurisdiction and entity deployment assumptions	Deployment form lists location, user type and activity.
GOV-11 [ ] NOT VERIFIED	Define independent validation and effective challenge process	Review criteria and reviewer independence recorded.
GOV-12 [ ] NOT VERIFIED	Maintain risk acceptance authority and expiration	Every material exception has approver and sunset date.
GOV-13 [ ] NOT VERIFIED	Define release change management and rollback authority	Signed change policy with authorized emergency approvers.
GOV-14 [ ] NOT VERIFIED	Create a no-go escalation process for material violations	Documented stop-work, notification and risk sign-off route.

ECO | Economy agent and ledger features (16 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
ECO-01 [ ] NOT VERIFIED	Implement typed economic state and immutable transitions	Property tests prevent mutation of prior snapshots.
ECO-02 [ ] NOT VERIFIED	Implement household assets, debt, income and consumption	Agent transitions preserve feasible household budgets.
ECO-03 [ ] NOT VERIFIED	Implement firm capital, wages, debt, output and inventory	Production and cash-flow balances reconcile under stress.
ECO-04 [ ] NOT VERIFIED	Implement bank loans, deposits, equity and reserves	Double-entry loan creation and repayment unit tests pass.
ECO-05 [ ] NOT VERIFIED	Implement central bank policy-rate rule baseline	Known reaction-function test scenarios reproduce coefficients.
ECO-06 [ ] NOT VERIFIED	Implement government taxes, transfers and expenditures	Sectoral flows reconcile to all counterpart accounts.
ECO-07 [ ] NOT VERIFIED	Define goods-market price and quantity clearing	Clearing residuals tracked with documented tolerance.
ECO-08 [ ] NOT VERIFIED	Define labor supply, demand and rationing	No double counting of hours or paid wages.
ECO-09 [ ] NOT VERIFIED	Define loan demand, underwriting and bank capital limits	Credit creation blocked on explicit constraint breaches.
ECO-10 [ ] NOT VERIFIED	Implement defaults and orderly loss allocation	Claims and equity loss reconciliation passes toy cases.
ECO-11 [ ] NOT VERIFIED	Implement interbank payments and reserve settlement	End-of-day account netting sums to zero by rule.
ECO-12 [ ] NOT VERIFIED	Implement deterministic heterogeneous-agent configuration	Same seed/config gives equivalent state and journal hash.
ECO-13 [ ] NOT VERIFIED	Implement observable-state windows for each agent type	No hidden future variables in policy feature vector.
ECO-14 [ ] NOT VERIFIED	Implement shocks to productivity, prices and monetary policy	Impulse response fixture reproduces expected sign/range.
ECO-15 [ ] NOT VERIFIED	Implement stylized-fact reports and macro aggregate outputs	GDP/inflation/credit definitions documented and validated.
ECO-16 [ ] NOT VERIFIED	Define shutdown behavior on broken accounting identity	Injected imbalance triggers fail-closed simulation halt.

MAR | Multi-agent reinforcement learning (14 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
MAR-01 [ ] NOT VERIFIED	Implement deterministic fixed-rule policy baselines	Baseline reward and economy trajectories independently reproducible.
MAR-02 [ ] NOT VERIFIED	Specify per-agent observation, action and reward spaces	Versioned schemas and economic rationale peer-reviewed.
MAR-03 [ ] NOT VERIFIED	Implement action feasibility projection/constraint solver	Property tests never produce prohibited spending or lending.
MAR-04 [ ] NOT VERIFIED	Implement household consumption/saving policy learning	Improvement checked against rule-based counterfactuals.
MAR-05 [ ] NOT VERIFIED	Implement firm production/investment policy learning	No negative output or investment outside feasible domain.
MAR-06 [ ] NOT VERIFIED	Implement bank underwriting and risk-limit agent	Credit risk and capital guardrails enforced in simulator.
MAR-07 [ ] NOT VERIFIED	Provide central-bank controller sandbox with policy constraints	Agent cannot issue actions outside allowed policy set.
MAR-08 [ ] NOT VERIFIED	Implement PPO or comparable baseline with deterministic seeding	Training manifest reproduces learning-curve statistics.
MAR-09 [ ] NOT VERIFIED	Measure multi-agent nonstationarity and instability	Report policy drift and convergence diagnostics across seeds.
MAR-10 [ ] NOT VERIFIED	Define reward hacking and degenerate agent safeguards	Adversarial tests attempt exploits in reward design.
MAR-11 [ ] NOT VERIFIED	Separate single-agent and co-learning experiments	Ablation isolates impact of simultaneous learning.
MAR-12 [ ] NOT VERIFIED	Measure welfare distribution, inequality and outcomes	Evaluation includes distributional and aggregate metrics.
MAR-13 [ ] NOT VERIFIED	Evaluate unseen tax/monetary parameter shifts	Held-out shock performance reported with uncertainty.
MAR-14 [ ] NOT VERIFIED	Document failure to converge and unidentifiable solutions	Negative result log maintained with triggering configurations.

LED | Financial accounting and market clearing (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
LED-01 [ ] NOT VERIFIED	Model every transaction as balanced journal entries	Debits equal credits across each currency and journal batch.
LED-02 [ ] NOT VERIFIED	Use deterministic integer minor units for cash ledger	No floating-cent drift in production ledger fixtures.
LED-03 [ ] NOT VERIFIED	Enforce nonnegative inventory unless specified financing exists	Invalid sale or withdrawal rejects with structured error.
LED-04 [ ] NOT VERIFIED	Track equity and P&L across all agent sectors	Equity rollforward matches realized changes and flows.
LED-05 [ ] NOT VERIFIED	Implement interest accrual and amortization schedule	Reference loan cash-flow tests match documented convention.
LED-06 [ ] NOT VERIFIED	Implement loan loss, write-off and recovery mechanics	Counterparty receivable/payable and equity reconcile.
LED-07 [ ] NOT VERIFIED	Implement corporate default and insolvency process	Loss waterfall tests pass prescribed creditor priority.
LED-08 [ ] NOT VERIFIED	Handle deposits, bank reserves, and interbank settlement	Cross-bank transfer preserves total settlement balances.
LED-09 [ ] NOT VERIFIED	Define market bid/ask spread and liquidity assumptions	No zero-cost trades unless explicit frictionless baseline.
LED-10 [ ] NOT VERIFIED	Track execution, settlement and position event order	Trade accounting never anticipates later settlement states.
LED-11 [ ] NOT VERIFIED	Implement time-zone, holiday, and business-day calendars	Weekend/holiday scenarios correctly advance timestamps.
LED-12 [ ] NOT VERIFIED	Prove global resource/currency conservation with tests	Randomized transaction sequences satisfy all invariants.

SDE | Classical stochastic finance (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
SDE-01 [ ] NOT VERIFIED	Implement GBM path generator with controlled Brownian increments	Empirical mean/variance match theory within CI.
SDE-02 [ ] NOT VERIFIED	Implement correlated multivariate asset simulator	Sample covariance and PSD checks pass.
SDE-03 [ ] NOT VERIFIED	Implement Heston price and variance dynamics	Compare moment/reference fixtures and nonnegative variance.
SDE-04 [ ] NOT VERIFIED	Implement variance discretization policy and error study	Method documented with convergence and positivity tests.
SDE-05 [ ] NOT VERIFIED	Implement numerical time-step refinement experiments	Path distribution stability as dt decreases is reported.
SDE-06 [ ] NOT VERIFIED	Implement regime-switching market baseline	Transition probabilities valid and stationary test available.
SDE-07 [ ] NOT VERIFIED	Implement volatility clustering benchmark where appropriate	Autocorrelation and conditional variance diagnostics recorded.
SDE-08 [ ] NOT VERIFIED	Implement jump or fat-tail stress scenario model	Parameterization and jump timing fully disclosed.
SDE-09 [ ] NOT VERIFIED	Distinguish physical P and risk-neutral Q model instances	Explicit measure field required in every engine call.
SDE-10 [ ] NOT VERIFIED	Implement historical calibration with dated parameter versions	Fit windows, optimizer state and calibration error logged.
SDE-11 [ ] NOT VERIFIED	Implement parameter constraints and identifiability diagnostics	Impossible values rejected; profile or sensitivity reported.
SDE-12 [ ] NOT VERIFIED	Test solver behavior on extreme but admissible inputs	No silent NaN, Inf or unbounded accepted paths.

NSD | Neural SDEs and learned market dynamics (13 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
NSD-01 [ ] NOT VERIFIED	Specify neural drift and diffusion architecture	Design record lists input features, units and constraints.
NSD-02 [ ] NOT VERIFIED	Implement differentiable stochastic integration path	Autograd gradient finite and numerically consistent on toy cases.
NSD-03 [ ] NOT VERIFIED	Enforce diffusion PSD/covariance construction	Every sampled covariance eigenvalue within numeric tolerance.
NSD-04 [ ] NOT VERIFIED	Control variance and price positivity by construction	Adversarial random inputs do not produce invalid states.
NSD-05 [ ] NOT VERIFIED	Implement regime-conditioned model with known-at time	Feature timestamps checked at every training/test event.
NSD-06 [ ] NOT VERIFIED	Implement train-only preprocessing and imputation	Validation/test never changes fitted normalization state.
NSD-07 [ ] NOT VERIFIED	Compare neural SDE with GBM/Heston and simple ML models	Pre-registered comparative benchmark report generated.
NSD-08 [ ] NOT VERIFIED	Measure distribution fit with proper scoring rules	CRPS/log score or justified alternatives reported.
NSD-09 [ ] NOT VERIFIED	Measure return tails, skew, volatility persistence	Moment diagnostics with confidence intervals available.
NSD-10 [ ] NOT VERIFIED	Implement multi-seed, multi-solver sensitivity study	Error bands across seeds and integration schemes documented.
NSD-11 [ ] NOT VERIFIED	Evaluate unseen macroeconomic shocks and drift	OOD risk vs baseline reported without re-tuning test.
NSD-12 [ ] NOT VERIFIED	Test exploding/vanishing gradients and numerical NaNs	Training automatically fails on loss/state integrity violations.
NSD-13 [ ] NOT VERIFIED	Register model version, training range and limitations	Reproducible model card with environmental dependencies.

PRI | Derivative instruments and pricing (13 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
PRI-01 [ ] NOT VERIFIED	Implement versioned instrument and payoff specifications	Independent payoff unit tests at boundaries and expiry.
PRI-02 [ ] NOT VERIFIED	Implement Black-Scholes European call and put	Analytical reference fixtures across strike and maturity grid.
PRI-03 [ ] NOT VERIFIED	Implement put-call parity and bounds verification	Property tests across rate, volatility, dividends if supported.
PRI-04 [ ] NOT VERIFIED	Implement Heston Fourier or characteristic-function pricer	Compare against high-precision published/reference values.
PRI-05 [ ] NOT VERIFIED	Implement risk-neutral calibration procedure	Implied-vol loss, parameter constraints and errors retained.
PRI-06 [ ] NOT VERIFIED	Implement Monte Carlo pricer with standard errors	Price estimates converge at predicted MC rate on fixtures.
PRI-07 [ ] NOT VERIFIED	Implement Delta/Gamma/Vega calculations with method tagging	Analytical/AD/finite-difference comparisons within tolerance.
PRI-08 [ ] NOT VERIFIED	Implement barrier option with discretization correction	Monitoring bias quantified against refined step benchmark.
PRI-09 [ ] NOT VERIFIED	Implement path-dependent Asian option with controls	Payoff and control-variate fixtures pass.
PRI-10 [ ] NOT VERIFIED	Implement portfolio-level repricing and Greeks aggregation	Sum of line-level Greeks reconciles to portfolio totals.
PRI-11 [ ] NOT VERIFIED	Validate volatility surface static-arbitrage conditions	Calendar and butterfly checks published with exceptions.
PRI-12 [ ] NOT VERIFIED	Fail closed on invalid parameters, expired instruments and bad units	Endpoint returns typed errors, not guessed prices.
PRI-13 [ ] NOT VERIFIED	Require numerical uncertainty with every nonclosed-form price	Response includes confidence/solver diagnostic and tolerance.

PDE | PINN and deep BSDE numerical solvers (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
PDE-01 [ ] NOT VERIFIED	Define PDE/BSDE, domain, terminal and boundary conditions	Model specification independently reviewed for correctness.
PDE-02 [ ] NOT VERIFIED	Implement classical grid/reference solver for small dimensions	Reference convergence tests on analytically solved examples.
PDE-03 [ ] NOT VERIFIED	Implement PINN interior residual loss and sampling scheme	Residual curves and sampling distribution reproducible.
PDE-04 [ ] NOT VERIFIED	Enforce payoff terminal condition during training	Terminal error grid checked and plotted.
PDE-05 [ ] NOT VERIFIED	Enforce applicable boundary and arbitrage constraints	Violation rates under stress below pre-approved tolerance.
PDE-06 [ ] NOT VERIFIED	Implement deep BSDE solver with causal time rollout	Numerical shape/gradient checks on known benchmark PDE.
PDE-07 [ ] NOT VERIFIED	Benchmark error by dimensions and number of time steps	Accuracy and complexity curves documented with hardware.
PDE-08 [ ] NOT VERIFIED	Benchmark price, Delta and Gamma errors separately	Greeks error does not exceed independent acceptance budget.
PDE-09 [ ] NOT VERIFIED	Audit near-expiry/at-the-money and discontinuous payoffs	Worst-case error surfaces explicitly included.
PDE-10 [ ] NOT VERIFIED	Perform optimizer and sampling ablations	Reproducible seeded matrix with independent validation.
PDE-11 [ ] NOT VERIFIED	Assess convergence failure and uncertainty	Solver refuses false precision when residuals remain high.
PDE-12 [ ] NOT VERIFIED	Compare neural compute cost against reliable numerical methods	CPU/GPU walltime, memory and accuracy reported fairly.

HDG | Deep hedging and stochastic control (14 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
HDG-01 [ ] NOT VERIFIED	Define portfolio wealth and terminal loss with cash financing	Self-financing or funding assumptions analytically documented.
HDG-02 [ ] NOT VERIFIED	Implement Black-Scholes model-consistent delta hedge	Reference P&L shape and convergence under zero frictions.
HDG-03 [ ] NOT VERIFIED	Implement cost-aware no-trade/threshold hedge	Transaction-cost comparison against delta hedge.
HDG-04 [ ] NOT VERIFIED	Implement MLP hedging policy with bounded actions	Training produces finite, feasible positions under stress.
HDG-05 [ ] NOT VERIFIED	Implement GRU/LSTM causal hedging policy	No future path information in recurrent observations.
HDG-06 [ ] NOT VERIFIED	Implement transaction costs and bid/ask execution model	Cost ledger reconciles to changes in holdings and cash.
HDG-07 [ ] NOT VERIFIED	Implement leverage, concentration and position constraints	All proposed actions obey projected feasible region.
HDG-08 [ ] NOT VERIFIED	Implement CVaR objective with correct loss definition	Quantile estimator checked against exact toy distribution.
HDG-09 [ ] NOT VERIFIED	Support stochastic optimization with detached test evaluation	Independent evaluation paths not used for tuning.
HDG-10 [ ] NOT VERIFIED	Implement probabilistic policy risk diagnostics	Gradient norms, action saturation and tail losses logged.
HDG-11 [ ] NOT VERIFIED	Evaluate out-of-family volatility and liquidity shocks	Risk report includes worse-than-baseline cases.
HDG-12 [ ] NOT VERIFIED	Report mean hedging P&L, CVaR, cost and turnover	Paired results show economic and statistical significance.
HDG-13 [ ] NOT VERIFIED	Perform reward hacking and unfair-baseline checks	Same information set, leverage bounds and execution costs.
HDG-14 [ ] NOT VERIFIED	Add controlled RL agent only after static policy passes	RL comparison includes training stability and risk bounds.

RISK | Risk metrics and factor analytics (13 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
RISK-01 [ ] NOT VERIFIED	Implement position normalization and pricing snapshot IDs	Every risk computation links exact holdings and market inputs.
RISK-02 [ ] NOT VERIFIED	Implement historical VaR with definitional metadata	Horizon, alpha, losses and interpolation declared.
RISK-03 [ ] NOT VERIFIED	Implement parametric VaR under explicit distribution	Distribution assumption and fit tests visible.
RISK-04 [ ] NOT VERIFIED	Implement Expected Shortfall with tie/atom-safe estimator	Reference discrete and continuous examples pass.
RISK-05 [ ] NOT VERIFIED	Implement volatility and covariance estimators	PSD and shrinkage safeguards unit-tested.
RISK-06 [ ] NOT VERIFIED	Implement historical drawdown and P&L attribution	Mark-to-market P&L sums to portfolio change.
RISK-07 [ ] NOT VERIFIED	Implement factor exposure estimation and residual risk	Exposure factor units and regression period tracked.
RISK-08 [ ] NOT VERIFIED	Implement VaR exception and coverage tests	Observed breaches compared with expected frequency and CI.
RISK-09 [ ] NOT VERIFIED	Implement ES forecasting evaluation and calibration diagnostics	Approved proper scoring/backtest methodology documented.
RISK-10 [ ] NOT VERIFIED	Implement economic stress scenario repricing	Every shock preserves parameter and scenario provenance.
RISK-11 [ ] NOT VERIFIED	Implement synthetic-bank network contagion risk	Synthetic exposures labeled and cascade conservation tested.
RISK-12 [ ] NOT VERIFIED	Implement risk-concentration and sensitivity decomposition	Aggregated components reconcile within specified method.
RISK-13 [ ] NOT VERIFIED	Implement risk report controls against stale market snapshots	Reject mismatched timestamps and missing model IDs.

OPT | Portfolio optimization and capital allocation (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
OPT-01 [ ] NOT VERIFIED	Implement budget-constrained mean-variance optimization	Optimality/feasibility checked against reference solver.
OPT-02 [ ] NOT VERIFIED	Implement max/min weight, leverage and short-sale controls	Solver refuses infeasible requests with diagnostic.
OPT-03 [ ] NOT VERIFIED	Implement CVaR portfolio optimization via scenario formulation	Toy exact solutions and tail behavior validated.
OPT-04 [ ] NOT VERIFIED	Implement turnover and proportional cost penalties	Cost tradeoff curves computed against naive solution.
OPT-05 [ ] NOT VERIFIED	Implement robust covariance/scenario uncertainty extension	Sensitivity to perturbation radii reported.
OPT-06 [ ] NOT VERIFIED	Implement liquidity and position-capacity constraints	No portfolio violates pre-approved participation limits.
OPT-07 [ ] NOT VERIFIED	Implement multi-period portfolio rebalancing simulator	Cash/position self-financing tests reconcile.
OPT-08 [ ] NOT VERIFIED	Include out-of-sample optimization comparison	Train-only parameter estimates; test is untouched.
OPT-09 [ ] NOT VERIFIED	Track solver name, tolerance, duality and termination state	No infeasible solution silently labeled optimal.
OPT-10 [ ] NOT VERIFIED	Estimate turnover-adjusted performance and drawdown	Metrics computed net of modeled cost and financing.
OPT-11 [ ] NOT VERIFIED	Stress portfolio against correlated crash regimes	Risk constraint violations and losses fully reported.
OPT-12 [ ] NOT VERIFIED	Implement explanation of allocation shifts and factor exposure	Report includes attributable economic drivers and residuals.

DAT | Data ingestion, licensing and provenance (14 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
DAT-01 [ ] NOT VERIFIED	Create inventory of each external data source and license	Approved source catalog with allowed use and redistribution.
DAT-02 [ ] NOT VERIFIED	Build FRED-MD/QD ingestion with metadata capture	Raw snapshots hashed; transforms and schema validated.
DAT-03 [ ] NOT VERIFIED	Build ALFRED/vintage-aware retrieval pipeline	Known-at views reproduce historical releases.
DAT-04 [ ] NOT VERIFIED	Build Treasury yield curve ingestion and conventions	Rate tenors, day count and holiday policies recorded.
DAT-05 [ ] NOT VERIFIED	Build Kenneth French factor ingestion with version tracking	Factors and methodological release notes versioned.
DAT-06 [ ] NOT VERIFIED	Implement raw/normalized/features/artifacts storage layers	Layer permissions and immutable boundaries tested.
DAT-07 [ ] NOT VERIFIED	Track observation, release, vintage and ingest timestamps	Data examples pass historical inclusion predicates.
DAT-08 [ ] NOT VERIFIED	Implement missing data and revision policy	No forward-fill across release windows without justification.
DAT-09 [ ] NOT VERIFIED	Enforce train/validation/test temporal partition on features	Automated leakage linter catches future timestamps.
DAT-10 [ ] NOT VERIFIED	Implement schema contracts, unit normalization and currency mapping	Contract failures quarantined and logged.
DAT-11 [ ] NOT VERIFIED	Monitor freshness, completeness, duplicates and revisions	Alerts and quality summary visible per dataset.
DAT-12 [ ] NOT VERIFIED	Hash normalized data and feature transformations	Every model reproduces dataset fingerprint exactly.
DAT-13 [ ] NOT VERIFIED	Prevent licensed/raw data from entering public release	CI secret/license scan and publication approval pass.
DAT-14 [ ] NOT VERIFIED	Document synthetic-vs-empirical labels and dataset cards	Every published result identifies data lineage and type.

EXP | Experiments, ablations and statistics (14 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
EXP-01 [ ] NOT VERIFIED	Pre-register primary endpoint and baseline comparison	Dated protocol stored before test runs.
EXP-02 [ ] NOT VERIFIED	Create factorial core economy/market/hedge experiment matrix	Eight primary combinations enumerated with seed schedule.
EXP-03 [ ] NOT VERIFIED	Specify random seed hierarchy across agent/market/training	Replay reconstructs nested RNG states and stream allocation.
EXP-04 [ ] NOT VERIFIED	Lock train/validation/test regimes and date cutoffs	Immutable split IDs and holdout manifest.
EXP-05 [ ] NOT VERIFIED	Add future-data negative control only for leakage diagnosis	Diagnostic clearly isolated from valid model scoreboard.
EXP-06 [ ] NOT VERIFIED	Include untrained and random-policy negative controls	Expected failure documented on reference scenario.
EXP-07 [ ] NOT VERIFIED	Compute paired confidence intervals for main comparison	Uncertainty method respects paired paths and dependence.
EXP-08 [ ] NOT VERIFIED	Specify multiple-testing treatment for secondary outcomes	Family and adjustment strategy recorded.
EXP-09 [ ] NOT VERIFIED	Support block bootstrap for serial dependence	Inference verified on correlated synthetic test cases.
EXP-10 [ ] NOT VERIFIED	Measure effect size and economic significance jointly	Improvement reported net of risk/cost definition.
EXP-11 [ ] NOT VERIFIED	Benchmark computational cost and memory across methods	Hardware, GPU driver, dtype and walltime logged.
EXP-12 [ ] NOT VERIFIED	Perform distribution shift and parameter perturbation tests	Predefined OOD scenarios yield separate scores.
EXP-13 [ ] NOT VERIFIED	Publish failure analyses and contradictory outcomes	No results excluded without transparent reason.
EXP-14 [ ] NOT VERIFIED	Require independent rerun of paper-quality claims	External/reviewer replication digest and comparison stored.

VAL | Numerical validation and software correctness (14 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
VAL-01 [ ] NOT VERIFIED	Define golden analytical test suite for GBM and Black-Scholes	Continuous integration matches known reference values.
VAL-02 [ ] NOT VERIFIED	Define MC sampling error test on benchmark payoffs	Expected confidence coverage across repeated simulations.
VAL-03 [ ] NOT VERIFIED	Define strong/weak convergence checks for stochastic integrators	Slope estimates within agreed theoretical/empirical range.
VAL-04 [ ] NOT VERIFIED	Validate covariance symmetry and positive semidefiniteness	Adversarial matrices rejected or repaired with warning.
VAL-05 [ ] NOT VERIFIED	Run automatic differentiation vs finite-difference Greek checks	Independent test grid matches within tolerance budget.
VAL-06 [ ] NOT VERIFIED	Validate Heston variance scheme across parameter extremes	No hidden negative values or silent clamps.
VAL-07 [ ] NOT VERIFIED	Check economic conservation under randomized transitions	Property tests preserve accounting and settlement identities.
VAL-08 [ ] NOT VERIFIED	Check budget, capital, leverage and action feasibility always	Invariant violation forces explicit fault state.
VAL-09 [ ] NOT VERIFIED	Implement deterministic CPU test mode with fixed seeds	Reference hashes match pinned version/environment.
VAL-10 [ ] NOT VERIFIED	Implement stochastic GPU tolerance-based reproducibility	Statistical equivalence defined where exact replay impossible.
VAL-11 [ ] NOT VERIFIED	Perform property-based and metamorphic test suites	Scaling, parity, relabeling and zero-shock tests documented.
VAL-12 [ ] NOT VERIFIED	Set numerical accuracy budget per API output	Tolerance fields present and tested for violations.
VAL-13 [ ] NOT VERIFIED	Perform NaN/Inf injection and solver fail-closed tests	No invalid output promoted or silently cached.
VAL-14 [ ] NOT VERIFIED	Cross-check pricing and ES with independent implementations	Two-method validation report and discrepancy resolution.

MLO | Training and model lifecycle (14 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
MLO-01 [ ] NOT VERIFIED	Define versioned training config for each model family	Committed config and strict schema validation.
MLO-02 [ ] NOT VERIFIED	Pin Python/PyTorch/library dependencies and environment	Lockfile and container digest rebuild successfully.
MLO-03 [ ] NOT VERIFIED	Track dataset hash, feature contract and training cutoffs	Model artifact includes unambiguous dataset provenance.
MLO-04 [ ] NOT VERIFIED	Persist training seeds, optimizer, scheduler and RNG state	Mid-run resume reproduces experiment within tolerance.
MLO-05 [ ] NOT VERIFIED	Track learning curves and gradient diagnostics	Experiment log contains step/epoch/seed metadata.
MLO-06 [ ] NOT VERIFIED	Implement early stopping using validation only	Test set never used for hyperparameter selection.
MLO-07 [ ] NOT VERIFIED	Version artifacts and keep immutable approved packages	Every model package resolves to unique content hash.
MLO-08 [ ] NOT VERIFIED	Create registry states: candidate/evaluated/approved/staged/retired	Transition policy enforced by code and audit trail.
MLO-09 [ ] NOT VERIFIED	Produce model cards for intended use and limitations	Card contains metric CIs, failure modes and risk owner.
MLO-10 [ ] NOT VERIFIED	Block deployment for missing provenance or validation	Promotion gate test denies incomplete candidates.
MLO-11 [ ] NOT VERIFIED	Implement reproducible inference container and input contracts	Same reference inputs produce approved outputs.
MLO-12 [ ] NOT VERIFIED	Implement controlled model rollbacks and retirement	Rollback test confirms restored model and stale-cache purge.
MLO-13 [ ] NOT VERIFIED	Track model drift, calibration and retraining triggers	Thresholds and escalation recorded before deployment.
MLO-14 [ ] NOT VERIFIED	Support comparison against last approved champion model	Champion/challenger report generated with matched data.

API | Service API and integration (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
API-01 [ ] NOT VERIFIED	Define versioned OpenAPI specification for service endpoints	Schema published with backward compatibility policy.
API-02 [ ] NOT VERIFIED	Validate input shape, timestamps, units and currencies	Invalid requests produce precise 4xx schema errors.
API-03 [ ] NOT VERIFIED	Implement async experiment creation and job status	Job IDs are unique, replayable and access controlled.
API-04 [ ] NOT VERIFIED	Implement compute quotas, cancellation and timeouts	Abusive or orphaned workloads terminate safely.
API-05 [ ] NOT VERIFIED	Define immutable pricing and risk report responses	Responses reference market snapshots and model versions.
API-06 [ ] NOT VERIFIED	Implement idempotency keys for job submission	Retrying create request does not duplicate a job.
API-07 [ ] NOT VERIFIED	Implement structured error codes and diagnostic IDs	Client can distinguish data, model and infra failures.
API-08 [ ] NOT VERIFIED	Implement cursor-based list and filtering for artifacts	Large collections paginate with stable ordering.
API-09 [ ] NOT VERIFIED	Implement service liveness and dependency readiness endpoints	Failed dependency removes readiness without crash loop.
API-10 [ ] NOT VERIFIED	Implement versioned typed SDK example	Client tests against API contract and compatibility fixtures.
API-11 [ ] NOT VERIFIED	Document trace IDs across job, inference and artifact storage	End-to-end request trace reconstructed in logs.
API-12 [ ] NOT VERIFIED	Exclude live trading, order execution and payment endpoints	Automated API surface scan enforces paper-only scope.

UIX | Dashboard, explainability and accessibility (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
UIX-01 [ ] NOT VERIFIED	Design research dashboard with economic simulation explorer	User can inspect macro shocks and aggregate time series.
UIX-02 [ ] NOT VERIFIED	Show agent/sector balance sheets and accounting residuals	Ledger reconciliation and filterable entries exposed.
UIX-03 [ ] NOT VERIFIED	Show market paths, volatility distributions and uncertainty	Confidence and simulation distributions clearly labeled.
UIX-04 [ ] NOT VERIFIED	Show derivative pricing diagnostics and Greeks	Value method, measure, model ID and as-of available.
UIX-05 [ ] NOT VERIFIED	Show hedge actions, realized costs and risk outcomes	Causal action timing and execution assumptions clear.
UIX-06 [ ] NOT VERIFIED	Show portfolio VaR/ES, factors, drawdown and stress	Horizon, units and confidence level displayed prominently.
UIX-07 [ ] NOT VERIFIED	Implement experiment comparison with CI and negative results	No cherry-picked score-only presentations by default.
UIX-08 [ ] NOT VERIFIED	Display synthetic-vs-historical labels on every visualization	Export and screenshots preserve provenance labels.
UIX-09 [ ] NOT VERIFIED	Implement audit trail view for model promotion and outputs	Reviewer can navigate report to evidence and approval.
UIX-10 [ ] NOT VERIFIED	Implement authorization-aware project sharing	Users cannot view unauthorized portfolios or experiments.
UIX-11 [ ] NOT VERIFIED	Create accessibility and responsive-layout test plan	Keyboard navigation and contrast review passes.
UIX-12 [ ] NOT VERIFIED	Implement report exports with reproducibility manifest	PDF/JSON export embeds run hash and method summary.

OPS | Observability and service operations (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
OPS-01 [ ] NOT VERIFIED	Create service inventory and dependency map	Every production component has owner and environment.
OPS-02 [ ] NOT VERIFIED	Emit structured JSON logs with sanitized fields	Sensitive data and secrets never logged in test scans.
OPS-03 [ ] NOT VERIFIED	Instrument distributed trace and run correlation identifiers	Request/job/model execution linked end to end.
OPS-04 [ ] NOT VERIFIED	Measure job queue depth, runtimes and cancellation count	Metrics dashboard with warning/critical thresholds.
OPS-05 [ ] NOT VERIFIED	Measure inference throughput, latency and failure rate	p50/p95/p99 trends shown per endpoint and model.
OPS-06 [ ] NOT VERIFIED	Measure data staleness and failed ingest counts	Alert exercise detects stale critical series.
OPS-07 [ ] NOT VERIFIED	Monitor unexpected NaN, solver failure and pricing deviation	High-severity numerical events page assigned owner.
OPS-08 [ ] NOT VERIFIED	Monitor model drift, risk metric calibration and tail breach	Model monitoring dashboard and investigation playbook.
OPS-09 [ ] NOT VERIFIED	Track GPU/CPU/RAM and artifact storage utilization	Capacity forecast and budget alerts configured.
OPS-10 [ ] NOT VERIFIED	Define actionable alert routing and severity taxonomy	Each P0/P1 alert maps to human owner and escalation.
OPS-11 [ ] NOT VERIFIED	Maintain runbooks for top failure categories	Simulated on-call drills complete with incident timeline.
OPS-12 [ ] NOT VERIFIED	Confirm no observability vendor exports restricted financial data	Data-flow review and redaction verification.

SEC | Security and software supply chain (15 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
SEC-01 [ ] NOT VERIFIED	Create threat model with assets, trust boundaries and misuse cases	Threat register reviewed and mapped to controls.
SEC-02 [ ] NOT VERIFIED	Require protected branches and peer-reviewed merges	Repository rules block unsigned/unreviewed changes.
SEC-03 [ ] NOT VERIFIED	Enforce MFA and least-privilege maintainers	Identity access review and access log evidence.
SEC-04 [ ] NOT VERIFIED	Implement SSO/OIDC and service identity at enterprise tier	Unauthorized requests denied, logged and tested.
SEC-05 [ ] NOT VERIFIED	Encrypt transit and storage with documented key ownership	TLS and at-rest configuration externally verified.
SEC-06 [ ] NOT VERIFIED	Manage secrets in vault, not code or CI logs	Secret scan and forced credential-rotation exercise pass.
SEC-07 [ ] NOT VERIFIED	Pin container base images and scan dependencies	SBOM and vulnerability reports attached to release.
SEC-08 [ ] NOT VERIFIED	Validate and sandbox untrusted uploaded dataset formats	Fuzz/test suite blocks unsafe deserialization paths.
SEC-09 [ ] NOT VERIFIED	Sign release artifacts and preserve checksum chain	Signature verification succeeds independently.
SEC-10 [ ] NOT VERIFIED	Implement network egress allowlist for ingestion and model serving	Unexpected egress denied and alert generated.
SEC-11 [ ] NOT VERIFIED	Rate-limit APIs and enforce per-tenant resource budgets	Load test demonstrates admission control under attack-like load.
SEC-12 [ ] NOT VERIFIED	Classify confidential portfolio and user data	Retention/handling/access labels audited by data owner.
SEC-13 [ ] NOT VERIFIED	Implement backup encryption and recovery access separation	Restored backups preserve authorization policy.
SEC-14 [ ] NOT VERIFIED	Perform independent penetration test before R2 release	Findings triaged; critical vulnerabilities closed.
SEC-15 [ ] NOT VERIFIED	Document vulnerability disclosure and patch cadence	Public security policy and response owner established.

PRV | Privacy, legal and data entitlements (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
PRV-01 [ ] NOT VERIFIED	Maintain dataset rights and vendor-entitlement register	Permitted uses, retention and redistribution independently checked.
PRV-02 [ ] NOT VERIFIED	Classify personal, financial and confidential records	Data processing inventory and purposes documented.
PRV-03 [ ] NOT VERIFIED	Minimize user and agent-associated personal information	Tests confirm default datasets avoid real PII.
PRV-04 [ ] NOT VERIFIED	Define data retention schedule for raw data, logs and artifacts	Automated expiry or signed exceptions verifiable.
PRV-05 [ ] NOT VERIFIED	Define deletion and data-subject request procedures where relevant	Dry-run removal traverses derived stores where required.
PRV-06 [ ] NOT VERIFIED	Review cross-border data transfer constraints per jurisdiction	Legal/data officer approval attached to environment.
PRV-07 [ ] NOT VERIFIED	Document licensing of source code, model weights and benchmarks	Chosen OSS license and third-party notice inventory.
PRV-08 [ ] NOT VERIFIED	Create external API/vendor contract security review	Service data flows and processors/permissions approved.
PRV-09 [ ] NOT VERIFIED	Define disclosure of simulated and financial data limitations	Public UI/export carries simulation/non-advice statement.
PRV-10 [ ] NOT VERIFIED	Check regulatory applicability before any client financial use	Counsel-signed scoping memo tied to actual jurisdiction.
PRV-11 [ ] NOT VERIFIED	Prohibit release of private benchmark results without consent	Publication redaction and entitlement verification pass.
PRV-12 [ ] NOT VERIFIED	Record approvals and scope for all data sharing	Access grant includes requester, owner, expiry and justification.

MGR | Independent model risk governance (14 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
MGR-01 [ ] NOT VERIFIED	Maintain enterprise model inventory with tier/materiality	Every active model has business purpose and owner.
MGR-02 [ ] NOT VERIFIED	Document conceptual soundness and theoretical assumptions	Independent reviewer evaluates mathematical logic.
MGR-03 [ ] NOT VERIFIED	Validate input data suitability and training provenance	Independent assessment reproduces eligibility and split.
MGR-04 [ ] NOT VERIFIED	Perform outcomes testing and backtests against baseline	Validation report covers calibration and operational use.
MGR-05 [ ] NOT VERIFIED	Document OOD boundaries and known failure modes	Model card lists tests, unresolved limitations and no-go cases.
MGR-06 [ ] NOT VERIFIED	Evaluate model dependency interactions and aggregated model risk	System-level error propagation analyzed, not just local scores.
MGR-07 [ ] NOT VERIFIED	Maintain effective challenge and reviewer independence	Sign-off reviewer has sufficient technical independence.
MGR-08 [ ] NOT VERIFIED	Require formal challenger benchmarking for material models	Approved reference, metrics, CI and failures recorded.
MGR-09 [ ] NOT VERIFIED	Document ongoing monitoring and recalibration thresholds	Active thresholds and actions approved before R2 deployment.
MGR-10 [ ] NOT VERIFIED	Maintain validation issue tracking and remediation	Every finding has severity, owner, expiry and disposition.
MGR-11 [ ] NOT VERIFIED	Restrict model usage to intended applications	Authorization and UI reject unsupported decisions.
MGR-12 [ ] NOT VERIFIED	Define model decommissioning and rollback procedure	Retired version not served; historical evidence preserved.
MGR-13 [ ] NOT VERIFIED	Review vendor and pretrained components where applicable	Third-party model limitations and validation recorded.
MGR-14 [ ] NOT VERIFIED	Ensure policy explicitly references relevant 2026 guidance	Legal/risk owner reviews SR 26-2 applicability/analogies.

CIQ | CI/CD, testing and artifact promotion (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
CIQ-01 [ ] NOT VERIFIED	Run lint, type check and formatting on pull requests	CI gate fails on style/type violations.
CIQ-02 [ ] NOT VERIFIED	Run fast unit/property/numerical regression tests	Main branch protected on successful test suite.
CIQ-03 [ ] NOT VERIFIED	Run deterministic reference experiment smoke test	Expected output hash or numeric tolerance matches.
CIQ-04 [ ] NOT VERIFIED	Run integration contract tests for service schemas	All endpoint clients pass backward compatibility suite.
CIQ-05 [ ] NOT VERIFIED	Run vulnerability, dependency and secret scans	High/critical findings block release unless approved.
CIQ-06 [ ] NOT VERIFIED	Build reproducible containers and create SBOM	Rebuild process and image digests archived.
CIQ-07 [ ] NOT VERIFIED	Sign artifacts and attach release provenance statement	Independent signature and source verification succeeds.
CIQ-08 [ ] NOT VERIFIED	Require independent model-validation gate before promotion	Two-person approval recorded in immutable system.
CIQ-09 [ ] NOT VERIFIED	Implement staging and canary/shadow deployment paths	Shadow comparison and rollback are exercised.
CIQ-10 [ ] NOT VERIFIED	Support automatic rollback on critical metric regression	Fault injection demonstrates reversal within recovery target.
CIQ-11 [ ] NOT VERIFIED	Store complete deploy manifests, config and feature flags	Any deployment can be recreated from recorded artifacts.
CIQ-12 [ ] NOT VERIFIED	Require maintenance window and notification for breaking change	Change log links affected clients and owner approval.

REL | Reliability, performance and capacity (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
REL-01 [ ] NOT VERIFIED	Document explicit internal service SLO/SLI definitions	Approved SLO sheet contains workload assumptions.
REL-02 [ ] NOT VERIFIED	Benchmark metadata API p95 against target on reference hardware	Load report includes request mix and concurrency.
REL-03 [ ] NOT VERIFIED	Benchmark simulation jobs at small/medium/large scales	Time/memory curves and resource budget published.
REL-04 [ ] NOT VERIFIED	Run concurrent simulation isolation and quota tests	No cross-job seed corruption or data leakage.
REL-05 [ ] NOT VERIFIED	Test graceful cancellation and partial job cleanup	Canceled jobs leave traceable terminal status and no lock.
REL-06 [ ] NOT VERIFIED	Test database/object storage failover for R2 scope	Fault exercise meets agreed recovery behavior.
REL-07 [ ] NOT VERIFIED	Perform rate limiting and backpressure at overload	System rejects work predictably rather than failing globally.
REL-08 [ ] NOT VERIFIED	Verify log/metric storage and cardinality budgets	Dashboard remains bounded during high run volume.
REL-09 [ ] NOT VERIFIED	Test long-horizon simulator stability and memory leaks	Extended soak test meets numerical and memory thresholds.
REL-10 [ ] NOT VERIFIED	Define capacity reservation and user quota policies	Requests exceeding quota rejected with transparent reason.
REL-11 [ ] NOT VERIFIED	Run backup restore and integrity validation	Restore artifacts are readable and hash-consistent.
REL-12 [ ] NOT VERIFIED	Record error budget actions and service ownership	Breaches trigger release freeze or corrective review.

DRI | Incidents, disaster recovery and business continuity (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
DRI-01 [ ] NOT VERIFIED	Define severity matrix for incorrect pricing, risk and data breaches	SEV tiers and response ownership approved.
DRI-02 [ ] NOT VERIFIED	Create numerical-integrity incident runbook	Incorrect output triggers isolate, invalidate and re-evaluate.
DRI-03 [ ] NOT VERIFIED	Create compromised credential and secret rotation runbook	Tabletop demonstrates revoked tokens and cleaned pipelines.
DRI-04 [ ] NOT VERIFIED	Create corrupt-data and point-in-time leak response	Affected runs traced, revoked and recomputed.
DRI-05 [ ] NOT VERIFIED	Document outage communications and stakeholder notification	Templates include impact, detection time and next steps.
DRI-06 [ ] NOT VERIFIED	Define incident commander, scribe and escalation rota	On-call process tested with realistic page.
DRI-07 [ ] NOT VERIFIED	Define backup RPO/RTO based on approved system tier	Recovery targets and evidence recorded.
DRI-08 [ ] NOT VERIFIED	Run restore drill using encrypted offsite snapshots	Full rebuild and reference risk report verified.
DRI-09 [ ] NOT VERIFIED	Run region/zone failure drill where applicable	Deployment-specific contingency fulfills defined RTO.
DRI-10 [ ] NOT VERIFIED	Create postmortem process with corrective action tracker	Blameless RCA and preventive test acceptance recorded.
DRI-11 [ ] NOT VERIFIED	Test audit-log availability throughout failover	Approval and model activity history remains complete.
DRI-12 [ ] NOT VERIFIED	Document decommission or safe-shutdown contingency	Unhealthy model service stops serving unapproved outputs.

AUD | Audit evidence, lineage and reporting (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
AUD-01 [ ] NOT VERIFIED	Assign immutable unique ID to each experimental run	Query resolves config, commit and all generated artifacts.
AUD-02 [ ] NOT VERIFIED	Attach data/version fingerprints to every result report	Report independently reconstructs input lineage.
AUD-03 [ ] NOT VERIFIED	Capture model training config, environment and RNG state	Stored manifest sufficient for replay at pinned environment.
AUD-04 [ ] NOT VERIFIED	Capture pricing measure and market snapshot for reports	Auditor can differentiate P risk model from Q pricer.
AUD-05 [ ] NOT VERIFIED	Capture agent and hedge decisions with timestamps	Action follows observed state in audit replay.
AUD-06 [ ] NOT VERIFIED	Capture all model registry and promotion transitions	Identity, approval, previous state and hash recorded.
AUD-07 [ ] NOT VERIFIED	Keep access-control changes and privileged actions logged	Regular audit review confirms tamper-evident retention.
AUD-08 [ ] NOT VERIFIED	Publish simulation versus empirical disclaimer in exports	Machine-readable source tag and visible watermark included.
AUD-09 [ ] NOT VERIFIED	Document reporting metric formulas, units and conventions	Each report can be recomputed from raw facts.
AUD-10 [ ] NOT VERIFIED	Version final analytical paper figures and tables	Figures link to reproducible scripts and data hashes.
AUD-11 [ ] NOT VERIFIED	Generate gate evidence bundle with manifest and reviewers	One-command export contains complete sign-off artifacts.
AUD-12 [ ] NOT VERIFIED	Retain failed experiment evidence, not just best outcomes	Result inventory contains exclusions and documented reasons.

CMP | Standards alignment and regulatory assessment (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
CMP-01 [ ] NOT VERIFIED	Map selected controls to NIST AI RMF Govern/Map/Measure/Manage	Applicability matrix cites specific outcomes; not faux certification.
CMP-02 [ ] NOT VERIFIED	Map software controls to NIST SSDF secure SDLC practices	Gap assessment and prioritized remediation recorded.
CMP-03 [ ] NOT VERIFIED	Map security outcomes to NIST CSF 2.0 as needed	Profile lists current/target state and evidence.
CMP-04 [ ] NOT VERIFIED	Evaluate 2026 revised interagency model-risk guidance applicability	Institution/legal reviewer documents scope and tailoring.
CMP-05 [ ] NOT VERIFIED	Map risk data lineage/report quality to BCBS 239 concepts	Reconciliation, completeness and timeliness tests available.
CMP-06 [ ] NOT VERIFIED	Assess whether EU DORA applies to actual deploying entity	Legal scoping decision and ICT third-party review recorded.
CMP-07 [ ] NOT VERIFIED	Assess regional personal-data law and data residency requirements	Counsel/privacy sign-off for chosen tenant jurisdiction.
CMP-08 [ ] NOT VERIFIED	Assess whether proposed use is research, advisory or regulated trading	Formal legal-use classification before client rollout.
CMP-09 [ ] NOT VERIFIED	If market access planned, conduct dedicated controls review	New project/approval for relevant market access safeguards.
CMP-10 [ ] NOT VERIFIED	Define third-party/vendor risk control catalogue	Due diligence, concentration and exit plans reviewed.
CMP-11 [ ] NOT VERIFIED	Define audit retention and recordkeeping by applicable scope	Retention schedule signed by records owner.
CMP-12 [ ] NOT VERIFIED	Obtain explicit disclaimer/claims review before external marketing	No statements implying statutory approval or guaranteed returns.

DOC | Developer documentation and handoff (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
DOC-01 [ ] NOT VERIFIED	Publish clear environment setup and one-command smoke test	Clean VM/container follows README successfully.
DOC-02 [ ] NOT VERIFIED	Publish research problem statement and falsifiable hypotheses	Protocol identifies primary endpoint, controls and scope.
DOC-03 [ ] NOT VERIFIED	Publish full mathematical notation and assumptions appendix	Symbols, units and integration conventions consistent.
DOC-04 [ ] NOT VERIFIED	Publish subsystems interfaces and simulation event-order spec	Architecture diagram and typed contracts available.
DOC-05 [ ] NOT VERIFIED	Publish data dictionary, feature definitions and known-at logic	Reader can reproduce a backdated feature row.
DOC-06 [ ] NOT VERIFIED	Publish model card template and validation documentation	Every model artifact includes intended use and limitations.
DOC-07 [ ] NOT VERIFIED	Publish experiment guide and seed/split protocol	Independent run matches selected report within tolerance.
DOC-08 [ ] NOT VERIFIED	Publish operational runbooks and rollback process	Operator tabletop passes dry-run actions.
DOC-09 [ ] NOT VERIFIED	Publish system security and data-handling guide	Principles and supported auth flows documented.
DOC-10 [ ] NOT VERIFIED	Publish API specification and example client	Client sends valid pricing/report requests in test environment.
DOC-11 [ ] NOT VERIFIED	Publish contributor guide, code of conduct and security policy	External contributor flow and disclosure mechanism tested.
DOC-12 [ ] NOT VERIFIED	Publish results including negative findings and limits	All cited metrics trace to run manifests and public assets.

LCH | Launch, user acceptance and release controls (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
LCH-01 [ ] NOT VERIFIED	Define release feature scope and hard exclusions	Signed checklist explicitly rejects order routing by default.
LCH-02 [ ] NOT VERIFIED	Freeze code and dependency lock for validation candidate	Candidate hash matches staging and acceptance tests.
LCH-03 [ ] NOT VERIFIED	Complete independent scientific/quantitative review	Open P0 model findings closed or formally blocked.
LCH-04 [ ] NOT VERIFIED	Complete security and privacy sign-off for defined R2 tier	No unresolved critical vulnerability or unauthorized data flow.
LCH-05 [ ] NOT VERIFIED	Complete operations readiness review and on-call assignment	Runbooks, alert tests, monitoring and escalation pass.
LCH-06 [ ] NOT VERIFIED	Complete disaster recovery restore drill and RTO evidence	Signed drill report meets scope-specific targets.
LCH-07 [ ] NOT VERIFIED	Complete load, soak and failure injection acceptance	Benchmark evidence attached to release decision.
LCH-08 [ ] NOT VERIFIED	Complete entitlements, licenses and publication review	No prohibited data or artifacts included in release.
LCH-09 [ ] NOT VERIFIED	Run controlled user acceptance tests for analysts and reviewers	Documented cases with approved pass/fail evidence.
LCH-10 [ ] NOT VERIFIED	Demonstrate versioned rollback and model invalidation	Canary rollback plus audit chain verified.
LCH-11 [ ] NOT VERIFIED	Approve external communications and limitation statements	Product wording clearly research/simulation where applicable.
LCH-12 [ ] NOT VERIFIED	Issue final GO/NO-GO decision with approver and conditions	Signed record references every required R2 gate and waiver.

EXT | Optional regulated/live-use extension: default NOT IN SCOPE (12 checks)

ID / status	Required capability or control	Acceptance evidence / pass condition
EXT-01 [ ] NOT VERIFIED	Propose separate legally reviewed product scope for execution	No change to default research API without new charter.
EXT-02 [ ] NOT VERIFIED	Obtain regulatory/legal applicability and licensing opinion	Named entity, venue, geography and activity documented.
EXT-03 [ ] NOT VERIFIED	Define broker/exchange connectivity and contractual obligations	Agreement, certification and authorized network path.
EXT-04 [ ] NOT VERIFIED	Define pre-trade credit, position and erroneous-order controls	Independent tests verify limits before an order is sent.
EXT-05 [ ] NOT VERIFIED	Implement market access permission separation and kill switch	Trading pathway defaults disabled and can be halted externally.
EXT-06 [ ] NOT VERIFIED	Implement real-time P&L, positions, and execution reconciliation	Independent broker statement and system positions reconcile.
EXT-07 [ ] NOT VERIFIED	Implement latency/failure handling and manual intervention	Fault injection proves controlled shutdown and operator escalation.
EXT-08 [ ] NOT VERIFIED	Implement trading surveillance, recordkeeping and retention	Legally appropriate controls reviewed by compliance.
EXT-09 [ ] NOT VERIFIED	Perform separate financial model validation for specific products	Valuation/risk outputs independently approved for intended use.
EXT-10 [ ] NOT VERIFIED	Conduct paper/shadow trading over institution-approved period	Live-like environment with no actual order placement.
EXT-11 [ ] NOT VERIFIED	Perform capital, liquidity and operational risk review	Limits and loss budgets signed by accountable officers.
EXT-12 [ ] NOT VERIFIED	Obtain multi-function final authorization for any live orders	Trading, risk, compliance, IT security and legal sign-off.

15 Readiness scorecard and completion arithmetic

This register has 374 individually testable checklist items in 29 categories. The optional regulated/live-trading track has 12 separately gated items, and the research/enterprise track has 362. All implementation checks remain NOT VERIFIED. Do not infer completed engineering from completed written requirements.

Group	Coverage	Checks	Verified	Gate
GOV	Program governance and scope	14	0	R1
ECO	Economy agent and ledger features	16	0	R1
MAR	Multi-agent reinforcement learning	14	0	R1
LED	Financial accounting and market clearing	12	0	R1
SDE	Classical stochastic finance	12	0	R1
NSD	Neural SDEs and learned market dynamics	13	0	R1
PRI	Derivative instruments and pricing	13	0	R1
PDE	PINN and deep BSDE numerical solvers	12	0	R1
HDG	Deep hedging and stochastic control	14	0	R1
RISK	Risk metrics and factor analytics	13	0	R1
OPT	Portfolio optimization and capital allocation	12	0	R1
DAT	Data ingestion, licensing and provenance	14	0	R1
EXP	Experiments, ablations and statistics	14	0	R1
VAL	Numerical validation and software correctness	14	0	R1
MLO	Training and model lifecycle	14	0	R1
API	Service API and integration	12	0	R1
UIX	Dashboard, explainability and accessibility	12	0	R1
OPS	Observability and service operations	12	0	R1
SEC	Security and software supply chain	15	0	R1
PRV	Privacy, legal and data entitlements	12	0	R1
MGR	Independent model risk governance	14	0	R1
CIQ	CI/CD, testing and artifact promotion	12	0	R1
REL	Reliability, performance and capacity	12	0	R1
DRI	Incidents, disaster recovery and business continuity	12	0	R1
AUD	Audit evidence, lineage and reporting	12	0	R1
CMP	Standards alignment and regulatory assessment	12	0	R1
DOC	Developer documentation and handoff	12	0	R1

Group	Coverage	Checks	Verified	Gate
LCH	Launch, user acceptance and release controls	12	0	R1
EXT	Optional regulated/live-use extension: default NOT IN SCOPE	12	0	R3

Progress formula and evidence rules

> Implementation verified coverage
>
> `Coverage = count(verified applicable checks) / count(applicable checks)`
>
> An N/A item is excluded only with written approval. Proposed or self-asserted completion does not count as verified.

Recommended issue tracker fields: check_id, gate, priority, status, owner, reviewer, opened_at, verified_at, evidence_uri, evidence_hash, build_sha, risk_exception, exception_expiry, notes. Evidence reviewers should be independent for material model risk/security/production-control items.

Release blocking rule

R0 needs correct invariants for the executed study. R1 needs a reproducible result chain, statistical protocol, held-out evaluation and documented limitations. R2 requires every scoped P0 control plus signed waivers for accepted noncritical gaps and a defined operational owner. R3 requires a separate activity-specific legal/regulatory and live-system approval; no check in this PDF substitutes for it.

16 Example evidence pack, templates and sign-offs

Experiment reproducibility manifest - required fields

```
run_id: pmth-exp-YYYYMMDD-XXXX
git_commit: <40-char SHA> image_digest: sha256:<digest>
dataset_vintage_hash: sha256:<digest> feature_contract_version: vN
splits: train/<id>, validation/<id>, test/<id>, heldout_shocks/<id>
seeds: {economy: N, market: N, initialization: N, evaluation: N}
math_model: {measure: P/Q, solver: ..., calendar: ..., units: ...}
metrics: {name: ..., horizon: ..., confidence: ..., scoring_rule: ...}
review: {researcher: ..., independent_validator: ..., date: ...}
results: {table_sha256: ..., plots_sha256: ..., limitations: ...}
```

Model card - mandatory sections

- Model identifier, type, authorship, training framework, checksum and license.
- Intended use, prohibited use, audience, supported geographies/asset universes.
- Input schema, required publication lag, market measure, feature transformations.
- Training dataset and vintage fingerprints, validation/test design and hyperparameters.
- Primary metrics, uncertainty intervals, calibration and subgroup/regime performance.
- Failure modes, numerical constraints, operational dependencies, expiration and monitoring.
- Independent reviewer, approval/exception record and rollback version.

Gate decision form

Field	To be populated at review
Gate / release	G0-G7; artifact version; proposed tenant and operating mode.
Scope	Enabled capabilities and specifically prohibited operations.
Evidence index	All required check IDs, run manifests, reviews and security artifacts.
Open findings	ID, severity, mitigation, date, accepted residual risk.
Scientific approval	Named reviewer, date, report identifier, caveats.
Security/ops approval	Named approvers, test runs, monitoring and incident ownership.
Legal/compliance assessment	Applicable only for chosen users, jurisdictions and activities.
Decision	GO / CONDITIONAL / NO-GO, timestamp, signers and sunset date.

Minimum v0.1 demonstration script

A reproducible one-command test should simulate a small closed economy, verify all journals, apply one known monetary shock, generate GBM/Heston market paths, price one European option under a separately calibrated Q model, compare delta against an initial cost-aware neural hedge, compute P&L and CVaR, and export an auditable report. If any invariant, valuation tolerance or data-timing check fails, the run must be marked invalid rather than silently published.

17 Risk register: what can invalidate the research

ID	Risk	Impact and severity	Mitigation
R-01	Economic invalidity Learning agents exploit accounting inconsistencies.	P0	Deterministic ledger invariants; constrained policy actions; reviewer.
R-02	Measure confusion P dynamics incorrectly used for risk-neutral derivatives prices.	P0	Explicit P/Q tags, pricing calibration and independent valuation.
R-03	Lookahead leakage Historical revisions appear in backdated forecasts.	P0	Release-time data view; negative control; lineage tests.
R-04	Simulation overfitting Neural hedge learns the simulator instead of robust risk control.	P0	Held-out market-model families and stress grids.
R-05	Numerical instability Heston/PINN paths explode or Greeks become unstable.	P0	Convergence grids, robust schemes and fail-closed outputs.
R-06	Reward hacking Central-bank/agent reward exploits loopholes.	P1	Welfare sensitivity, rule baselines, adversarial scenarios.
R-07	Inadequate data Sparse option surfaces or confidential exposures unavailable.	P1	Synthetic vs empirical labels; licensed vendor plan.
R-08	Nonreproducibility GPU or stochastic seeds produce irreconcilable claims.	P0	Manifest, statistical tolerance, controlled re-execution.
R-09	Financial misuse Simulation reports mistaken for live-trading advice.	P0	Access control, prominent disclaimer, disabled execution.
R-10	Privacy / license breach Vendor or financial data leaked via repo or telemetry.	P0	Entitlement map, scan, egress controls, audit.
R-11	Model governance gap Unvalidated candidate used in enterprise decision workflow.	P0	Registry gate, effective challenge, approved usage.
R-12	Operational unavailability Research service unavailable during critical review.	P1	Measured SLO, backup, incident response, DR test.
R-13	False significance Hyperparameter sweep leads to overstated findings.	P0	Pre-registration, multiple comparison plan, holdout.
R-14	Cost runaway MARL/stress-grid sweep exceeds compute budget.	P1	Resource caps, profiling and baseline-first roadmap.
R-15	Regulatory misinterpretation Framework named as certification or universal legal duty.	P0	Legal applicability memo and claims approval.

18 Proposed research deliverables and repository releases

Deliverable inventory

Artifact	Minimum contents	Exit criterion
Paper A - Economic kernel	Stock-flow rules, MARL, shock response, failure modes	Economy invariants and held-out shocks replicated.
Paper B - Stochastic hedging	Heston baseline, neural SDE, hedge objective, OOD	Pre-registered loss scores and multiple seeds.
Paper C - Integrated system	System feedback experiments and risk spillovers	Independent validation and ablation credibility.
Open-source release	Installable package, tests, examples and docs	One-command reproducible demo; license audit.
Benchmark suite	Reference datasets/scenarios, splits and scoring	Immutable test protocol; independent rerun.
Model and dataset cards	Intended use, limitations, provenance and metrics	Every shared artifact has approved card.
Enterprise evidence pack	Threat model, validation, runbooks, DR, SLO	R2 gate decision signed for defined tenant/scope.

Reference repository layout

```
prometheus-econ/
configs/{economy,markets,pricing,hedging,risk,experiments}/
src/prometheus/{core,econosim,stochastica,volterra,titan,api}/
src/prometheus/{data,models,evaluation,security,reporting}/
tests/{unit,property,integration,numerical,security,performance}/
experiments/{macro_feedback,neural_sde,hedging,pricing,systemic}/
dashboards/ notebooks/ docs/ research/ benchmarks/
infra/{local,staging,enterprise}/ .github/workflows/
pyproject.toml uv.lock README.md SECURITY.md LICENSE
```

Dependency choices

Core numeric: NumPy, SciPy, pandas/Polars, statsmodels, CVXPY. Learning: PyTorch, and a carefully chosen MARL/PINN framework only when a justified advantage exists. Storage: Parquet/DuckDB locally and Postgres/object store for internal deployment. Interface: Pydantic and FastAPI for typed research APIs. Quality: pytest, hypothesis, mypy/pyright, ruff, pre-commit, coverage. Reproducibility: pinned lock file, OCI image digest, deterministic seeded harness and artifact manifest.

19 Industrial readiness: final release decision

This page is the high-level GO/NO-GO checklist for an internal enterprise research analytics deployment. It is deliberately stricter than an academic GitHub demo, and it is not authorization to serve real-time execution or regulated advice.

Control gate	Required acceptance evidence	Current status
Scientific validity	Independent replication, documented limitations, no P0 math/data issues	NOT VERIFIED
Data governance	Point-in-time lineage, licensing, confidential-data controls	NOT VERIFIED
Model risk	Independent soundness/outcomes review and bounded intended use	NOT VERIFIED
Security	Identity, secrets, scanning, threat review and penetration test	NOT VERIFIED
Reliability	Measured SLO/load/soak tests and fault-safe numeric behavior	NOT VERIFIED
Operations	Alerts, on-call ownership, runbooks, DR restore and rollback	NOT VERIFIED
Audit	Full run-to-report lineage, tamper-evident approvals and retention	NOT VERIFIED
Legal	Entity/jurisdiction/use-case specific approval, truthful claims	NOT VERIFIED
Release management	Canary, two-person approval, rollback and change notice	NOT VERIFIED
Product scope	Research-only interfaces; no unapproved live execution path	NOT VERIFIED

CURRENT SYSTEM DECISION: NO-GO FOR PRODUCTION
The PDF is complete as a design, feature catalog and audit-ready checklist. No implemented system, trained model, datasets, acceptance tests, security controls, operational drills or deployment authorization have been evidenced. Therefore the project is not yet production-ready. First target: G1 economic ledger correctness, then G2 pricing benchmarks and G3 deep hedging.

Sign-off slots (unfilled)

Scientific lead: _____ Date: _____ Decision: _____

Independent model validation: _____ Date: _____ Decision: _____

Security / privacy: _____ Date: _____ Decision: _____

SRE / operations: _____ Date: _____ Decision: _____

Legal / compliance (if applicable): _____ Date: _____ Decision: _____

Final accountable release owner: _____ Date: _____ GO / NO-GO: _____

20 Selected bibliography and standards

Primary references and official guidance below are included to ground the design. The project does not claim validation or certification by any cited organization. Checked for this specification on 02 November 2026.

[R01] Buehler, Gonon, Teichmann & Wood (2019). Deep hedging. Quantitative Finance.
https://doi.org/10.1080/14697688.2019.1571683

[R02] Heston (1993). A closed-form solution for options with stochastic volatility. Review of Financial Studies.
https://doi.org/10.1093/rfs/6.2.327

[R03] Han, Jentzen & E (2018). Solving high-dimensional PDEs using deep learning. PNAS.
https://doi.org/10.1073/pnas.1718942115

[R04] Federal Reserve Bank of St. Louis. FRED-MD and FRED-QD databases.
https://www.stlouisfed.org/research/economists/mccracken/fred-databases

[R05] Federal Reserve Bank of St. Louis. ALFRED realtime/vintage API concepts.
https://fred.stlouisfed.org/docs/api/fred/realtime_period.html

[R06] US Department of the Treasury. Daily Treasury interest rates data.
https://home.treasury.gov/resource-center/data-chart-center/interest-rates

[R07] Kenneth R. French Data Library. Asset pricing factor returns.
https://mba.tuck.dartmouth.edu/pages/faculty/Ken.French/data_library.html

[R08] NIST. AI Risk Management Framework (AI RMF) 1.0 and Playbook; voluntary.
https://www.nist.gov/itl/ai-risk-management-framework

[R09] NIST SP 800-218. Secure Software Development Framework (SSDF), v1.1.
https://csrc.nist.gov/pubs/sp/800/218/final

[R10] NIST. Cybersecurity Framework 2.0.
https://www.nist.gov/cyberframework

[R11] US Federal Reserve. SR 26-2 (17 Apr 2026), Revised Guidance on Model Risk Management.
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[R16] Zheng et al. (2022). The AI Economist: taxation policy design via two-level deep MARL. Science Advances.
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[R17] PFHedge. Open-source PyTorch framework for deep hedging.
https://github.com/pfnet-research/pfhedge

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[R19] NIST AI RMF Playbook: Govern, Map, Measure, Manage.
https://airc.nist.gov/airmf-resources/playbook/

[R20] Federal Reserve SR 26-2 guidance full text landing page.
https://www.federalreserve.gov/frrs/guidance/supervisory-guidance-on-model-risk-management.htm

Appendix A Definition of done and issue example

Done = tested, evidenced, reviewed, scoped

A task is NOT done merely because code exists, a notebook executes, a model trains, or a graph looks plausible. The associated test must exercise its acceptance criteria, an artifact must be immutable and discoverable, its known limitations must be documented, and an appropriate reviewer must accept the evidence. Critical exceptions require explicit approval with expiry.

Example acceptance record - intentionally not completed

```
check_id: HDG-08
requirement: CVaR objective with correct terminal hedging loss
status: NOT VERIFIED
owner: unassigned reviewer: unassigned
acceptance_tests: discrete quantile toy; differentiability; loss convention
evidence_uri: null evidence_hash: null
verified_at: null risk_exception: null
next_action: implement reference estimator and independent tests
```

Appendix B Suggested research scenario catalog

Scenario	Shock	Required outputs
SC-01	Unanticipated +200 bp policy-rate shock	Rate path; borrower costs; bond repricing; hedge costs.
SC-02	Persistent inflation shock	Real consumption, rates, asset correlations, ES.
SC-03	Credit supply contraction	Loan rationing, firm output, insolvencies, contagion.
SC-04	Household employment collapse	Demand, nonperforming loans, balance-sheet stress.
SC-05	Productivity expansion	Wages, output, prices and financial valuations.
SC-06	Volatility regime break	Heston/neural SDE fit and hedge OOD stability.
SC-07	Funding liquidity freeze	Spread, margin, hedging frequency and execution failure.
SC-08	Correlation convergence crisis	Diversification failure and tail portfolio loss.
SC-09	Simulated bank default chain	Network exposure contagion and aggregate credit contraction.
SC-10	Macro-statistics release revision	Point-in-time retrieval and historical forecast consistency.
SC-11	Pricing surface arbitrage perturbation	Volatility surface validator rejection/correction.
SC-12	Outlier and malformed price event	Schema validation and fail-closed numerical pipeline.
SC-13	Economic agent reward exploit	Learned rule manipulation and invariant protection.
SC-14	Model distribution shift after deployment	Drift alerts, shadow baseline and rollback decisions.
SC-15	Infrastructure storage outage	Checkpoint integrity, audit retention and restoration.
SC-16	Insufficient market data coverage	Uncertainty escalation and abstention from price claim.

Appendix C Formal next actions

Priority 1 is not building the Transformer. It is making a constrained, correctly accounted toy economy and a trustworthy valuation/risk baseline. The proposed first pull requests are:

PR	Name	Minimum scope
PR-001	Repository foundations	Package, linting, CI, seed management, typed config, licensing.
PR-002	Stock-flow ledger	Double-entry journals, household/firm/bank objects, invariant tests.
PR-003	Macro time engine	Known-at releases, event schedule, deterministic shock controls.
PR-004	Stochastic reference	GBM and Heston path solvers with numerical benchmark tests.
PR-005	Pricing reference	Black-Scholes, Greeks, option interface, MC comparison.
PR-006	Risk reference	PnL, historical VaR and ES, loss estimators, report schema.
PR-007	Deep hedge v0	MLP hedging policy and cost-aware delta control.
PR-008	First research benchmark	Held-out seeds, cost accounting, bootstrap CI, reproducer.
PR-009	Macro-financial link	Policy-rate scenario mapped into market conditional states.
PR-010	Evidence export	Run manifest, model card, simulation disclaimers, gate report.

End of master specification. Complete the implementation evidence, independently review the scientific claims, and record sign-offs before changing any NOT VERIFIED status to VERIFIED. The correct initial production verdict remains NO-GO.